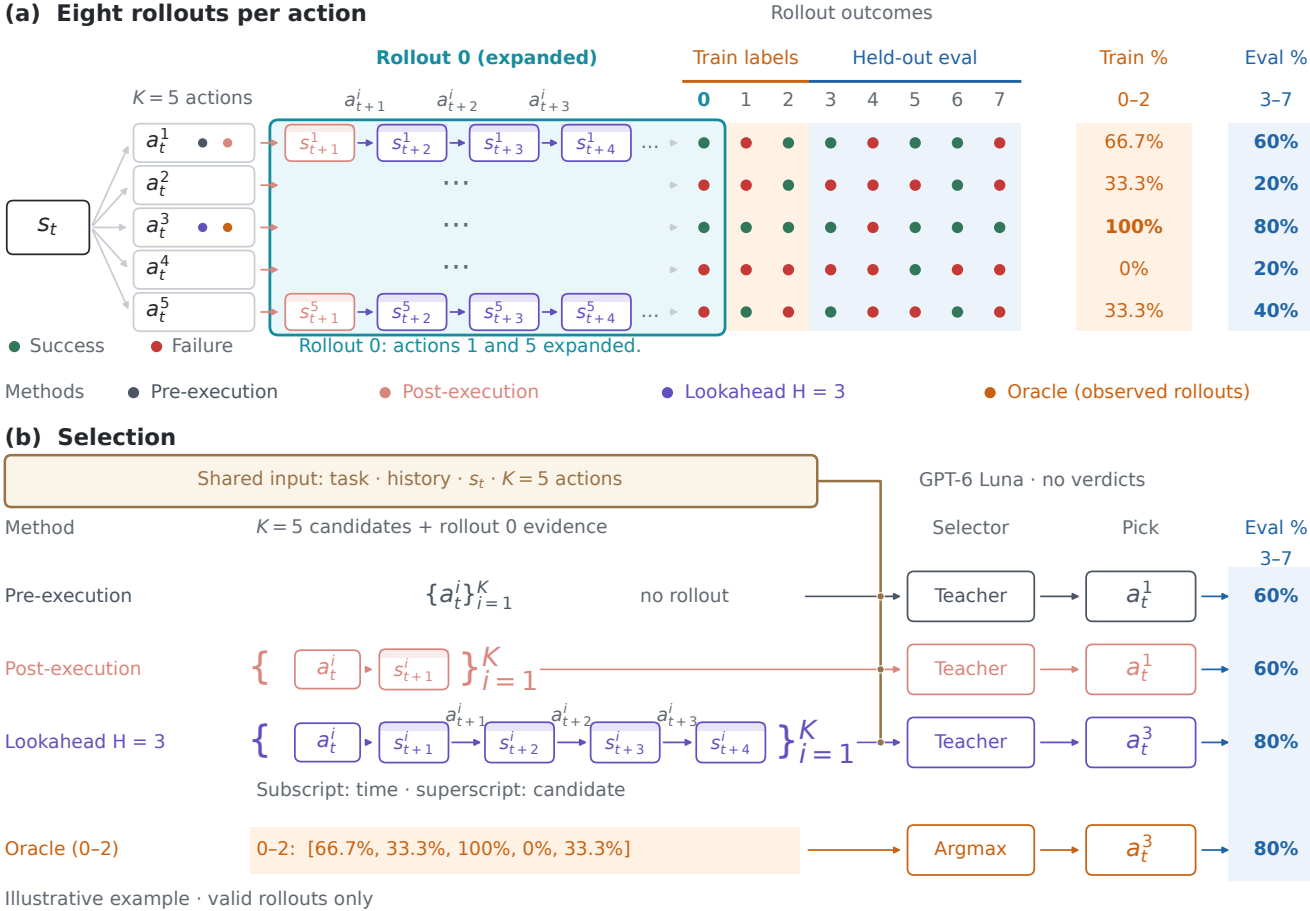
(a) Eight rollouts per action
Rollout outcomes
Rollout 0 (expanded)
Train labels
Held-out eval
Train %
Eval %
K = 5 actions
0 1 2 3 4 5 6 7
0–2
3–7
66.7%
60%
33.3%
20%
100%
80%
0%
20%
33.3%
40%
Success
Failure
Rollout 0: actions 1 and 5 expanded.
Methods
Pre-execution
Post-execution
Lookahead H = 3
Oracle (observed rollouts)
(b) Selection
Shared input: task · history · K = 5 actions
GPT-6 Luna · no verdicts
Method
K = 5 candidates + rollout 0 evidence
Selector
Pick
Eval %
3–7
Pre-execution
no rollout
Teacher
60%
Post-execution
Teacher
60%
Lookahead H = 3
Teacher
80%
Subscript: time · superscript: candidate
Oracle (0–2)
0–2: [66.7%, 33.3%, 100%, 0%, 33.3%]
Argmax
80%
Illustrative example · valid rollouts only